

Section A – NumPy Practical Questions

NumPy Basics & Array Creation

1. Create a NumPy array from 1 to 20 and print only even numbers using slicing.
2. Create a 3D NumPy array of shape (2,3,4) using np.arange() and print its dimensions.
3. Write a program to create arrays using zeros(), ones(), full(), and eye() functions.
4. Create two arrays and perform element-wise addition, subtraction, multiplication, and division.
5. Generate a random array of 15 elements between 10 and 100 and print maximum and minimum values.
6. Create a NumPy array and convert its datatype from integer to float and float to string.
7. Create a matrix of shape (4,4) and replace all diagonal values with 0.
8. Generate a random integer array and count how many elements are greater than 50.
9. Create a 1D array and convert it into:
 - Row vector
 - Column vector
 - 3D array
10. Write a program to compare memory usage between Python list and NumPy array.

NumPy Indexing & Slicing

11. Create a 5x5 matrix and print:
 - First row
 - Last column
 - Middle 3x3 matrix
 - Alternate rows
12. Create a 2D array and print all corner elements using indexing.
13. Write a program to reverse a NumPy array using slicing.
14. Create a 3D array and access:
 - First block
 - Last row of second block
 - Specific element using multi-dimensional indexing
15. Create a matrix and replace all even indexed rows with zero.
16. Extract all odd numbers from an array without using loops.

17. Create a 4x4 matrix and swap first and last columns.
18. Write a program to find positions of elements greater than 40 using `np.where()`.
19. Create a random array and sort only the first half of the array.
20. Create two arrays and print common elements between them.

Shape, Reshape & Iteration

21. Create a NumPy array of 24 elements and reshape it into:
 - (2,12)
 - (3,8)
 - (2,3,4)
22. Write a program to flatten a 3D array using:
 - `flatten()`
 - `ravel()`
23. Create a matrix and iterate:
 - Row-wise
 - Column-wise
 - Element-wise using `nditer`
24. Use `ndenumerate()` to print index and value of every element.
25. Create two matrices and perform horizontal stacking and vertical stacking.
26. Split a NumPy array into 4 equal parts using `array_split()`.
27. Write a program to concatenate two 2D arrays row-wise and column-wise.
28. Create a matrix and reshape it into another valid dimension without changing data.
29. Generate a random matrix and calculate:
 - Sum of each row
 - Mean of each column
 - Standard deviation of all elements
30. Create a 3D array and print all blocks separately using loops.

Searching, Sorting & Random Functions

31. Create a random integer array and sort it in descending order.
32. Write a program using `np.searchsorted()` to insert elements into sorted arrays.
33. Generate random numbers using:

- `rand()`
- `randint()`
- `choice()`
- `normal()`

34. Simulate rolling a dice 100 times using NumPy random functions.

35. Generate a random dataset of student marks and calculate:

- Highest marks
- Lowest marks
- Average marks
- Students scoring above average

36. Create a matrix and sort values based on a specific column.

37. Write a program to shuffle elements of an array.

38. Create a random array and find unique elements with their counts.

39. Create two matrices and perform matrix multiplication using NumPy.

40. Create a mini project:

- Generate random sales data for 12 months
- Calculate yearly total
- Find best sales month
- Plot sales trend using NumPy and Matplotlib

Section B – Pandas Practical Questions

Pandas Basics

41. Create a Pandas Series using a Python list and dictionary.
42. Create a DataFrame containing student information and display first and last 3 rows.
43. Import a CSV dataset and display:
 - Shape
 - Columns
 - Datatypes
 - Statistical summary
44. Create a DataFrame and add a new column called "Total Marks".
45. Write a program to rename columns in a DataFrame.
46. Create a DataFrame and remove duplicate rows.
47. Write a program to check null values and fill them using mean and median.
48. Filter students whose marks are greater than 70 using boolean indexing.
49. Sort a DataFrame based on salary column in descending order.
50. Create a DataFrame and apply conditional formatting logic using `apply()`.

Data Analysis & Manipulation

51. Create two DataFrames and perform:
 - Merge
 - Concatenate
 - Join
52. Group employee data department-wise and calculate average salary.
53. Create a dataset and find:
 - Maximum value
 - Minimum value
 - Mean
 - Median
 - Mode
54. Create a DataFrame containing product sales and identify top 5 selling products.
55. Use `value_counts()` to analyze frequency of categorical data.
56. Create a dataset and export it to CSV and Excel formats.

57. Read a dataset and drop unnecessary columns.

58. Create a DataFrame and convert categorical values into numerical values.

59. Build a mini analysis project using Pandas:

- Create employee dataset
- Calculate bonuses
- Filter highest performers
- Save final report into CSV

60. Create a practical project:

- Student result management system using Pandas
- Add grades based on marks
- Calculate percentage
- Find topper of class

Section C – Matplotlib Practical Questions

61. Create a line plot showing monthly sales data.
62. Plot multiple lines in a single graph with legends and labels.
63. Create a bar chart comparing marks of different students.
64. Create a horizontal bar graph for product sales.
65. Generate a pie chart showing market share of mobile brands.
66. Create a histogram for randomly generated marks distribution.
67. Plot scatter graph comparing study hours and marks.
68. Customize graph using:
 - Title
 - Grid
 - Color
 - Marker
 - Line style
69. Create subplots displaying:
 - Line chart
 - Bar chart
 - Pie chart
70. Create a graph showing temperature changes for 7 days.
71. Build a mini project:
 - Analyze company profit for 12 months
 - Plot line chart and bar chart
 - Highlight highest profit month
72. Create a graph comparing performance of 3 students in 5 subjects.

Section D – Seaborn Practical Questions

73. Load an inbuilt Seaborn dataset and display basic information.
74. Create a heatmap for a correlation matrix.
75. Generate a pairplot for a student performance dataset.
76. Create a countplot for categorical data analysis.
77. Plot a boxplot to detect outliers in salary data.

78. Create a violin plot comparing marks of students from different classes.
79. Generate a histogram with KDE using Seaborn.
80. Create a scatterplot comparing age and income.
81. Create a barplot showing average marks subject-wise.
82. Build a mini visualization dashboard using Seaborn:
 - Heatmap
 - Countplot
 - Scatterplot
 - Histogram

Advanced Practical & Coding-Based Questions

83. Create a NumPy-based attendance management system storing attendance in matrix form.
84. Build a Pandas-based cricket score analysis project.
85. Create a sales prediction visualization using Matplotlib and random generated data.
86. Use Seaborn to analyze Titanic dataset and identify survival trends.
87. Create a NumPy matrix calculator supporting:
 - Addition
 - Subtraction
 - Multiplication
 - Transpose
88. Build a Data Analysis project combining NumPy + Pandas + Matplotlib.
89. Generate random student marks using NumPy and visualize grade distribution using Seaborn.
90. Create a complete Employee Management Analysis project using Pandas and Matplotlib.
91. Create a weather data analysis project using random generated temperatures.
92. Build a COVID-style dataset analysis using Pandas and Seaborn.
93. Create a stock market visualization using Matplotlib with random stock prices.
94. Write a program to compare execution speed between Python lists and NumPy arrays.
95. Create a dashboard showing monthly expenses using Pandas and Matplotlib.
96. Create a heatmap for student marks using Seaborn.
97. Build a movie rating analysis project using Pandas.
98. Create a NumPy simulation of lottery number generation.
99. Create a Seaborn visualization project analyzing car mileage dataset.

100. Build a final integrated mini project using:

- NumPy for data generation
- Pandas for analysis
- Matplotlib for plotting
- Seaborn for advanced visualization